

Apex Gym Hub

SaaS Project Specification & Technical Documentation

Prepared for: uniyalmanas / wedd-builder

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Deployment URL: manas-portfolio-seven.vercel.app/gym-app/

GitHub Repository: github.com/uniyalmanas/GYM-Builder

System Version: v1.0.0 (Production Ready)

Apex Gym Hub SaaS Project Specification

A premium, production-grade Software-as-a-Service (SaaS) operational core and interactive member dashboard designed for elite fitness centers, athletic clubs, and personal training coaches.



User Interface Mockups

1. Premium Client Dashboard

The client interface is styled with a sleek dark theme and emerald accents, prioritizing user visual engagement. It features an interactive workout presets logger, a consistency circular meter, and a class/trainer appointment scheduler.



Apex Gym Hub



Apex Gym Hub

Dashboard

Workouts

Nutrition

Progress

Members

Settings

Welcome back, Alex Chen!

Tuesday, October 26, 2023



Quick actions



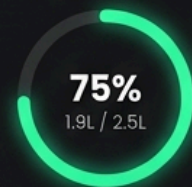
ACTIVE WORKOUT SCHEDULES

- [10:00 AM] ● High-Intensity Interval (H.I.I.T)
Sarah J. (60 mins)
- [5:30 PM] ● Power Lifting
Mike R. (45 mins)
- [7:00 PM] ● Recovery Yoga (Virtual)

DAILY WATER LOGGING



[Water Intake]
75% | 1.9L / 2.5L



+250ml

+500ml

+750ml

CONSISTENCY SCORE



QUICK STATS GRID

[Weight Progress]

82.5kg

(-0.8kg this week)

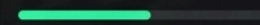


[Active Membership]

'ELITE TIER'

Valid until Oct 2024

Guest Passes: 4 Left



[Classes This Week]

9 / 10



[PRs Set]

3

Deadlift, Bench Press



2. CRM & Operator Console

The administrative control deck allows gym staff to manage user accounts, change subscription plans, edit system roles inline, track inbound leads, and adjust pricing parameters graphically.

[Dashboard](#)[Members](#)[Staff](#)[Classes](#)**Alex R.**
Gym Manager

REGISTER NEW MEMBER

Full Name

Sarah J. Adams

Email

sarah.adams@email.com

Phone

123-456-7890

Plan

Premium Monthly

Join Date

2023-10-27

Notes

Active runner...

Register Member

ACTIVE GYM MEMBERS (218 Total)

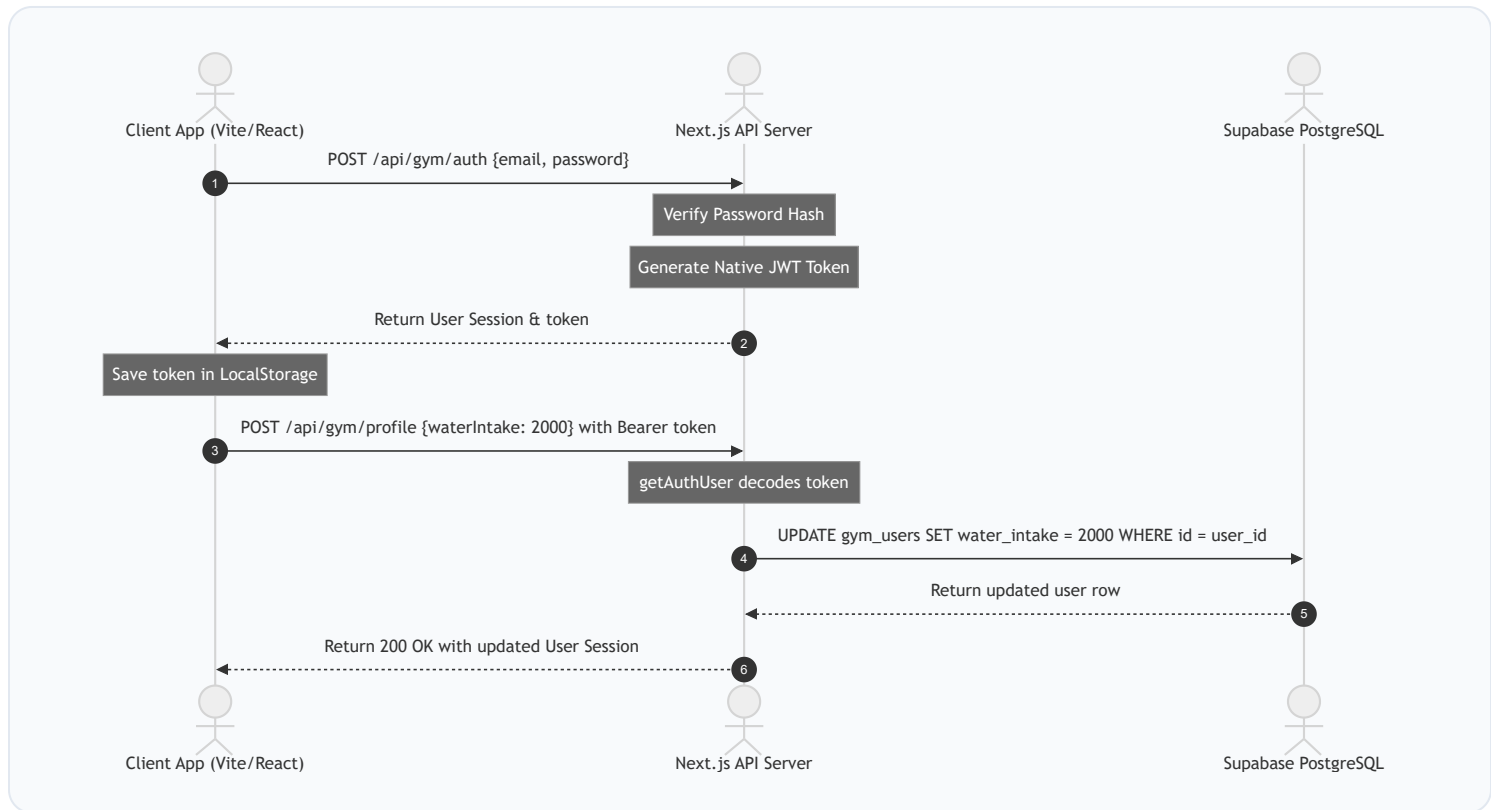
Name	System Role	Email	Joined	Last Active	Actions
 Emily Chen	ADMIN	emily.c@gym.com	2023-01-15	Today 09:12	 
 David Kim	ADMIN	david.k@gym.com	2022-06-10	Today 11:30	 
 Chloe Brown	MEMBER	chloe.b@email.co	2023-05-22	Today 08:45	 
 Michael Rodriguez	TRAINER	m.rod@email.com	2023-09-01	Oct 25	 
 Michael Rodriguez	MEMBER	m.rod@email.com	2023-09-01	Oct 25	 
 Sarah Johnson	TRAINER	sarah.j@email.con	2023-09-01	Oct 25	 
 Sarah Johnson	TRAINER	sarah.j@email.con	2022-11-20	Today 10:05	 
 Noah Williams	MEMBER	noah.w@gym.com	2023-10-10	Oct 26	 

218 Active Members

< 1 >

System Architecture & Data Flow

Below is a sequence diagram showcasing how a user logs in, receives a token, and makes authenticated API requests to update their daily hydration log.



Database Schemas

The application connects to a cloud-hosted Supabase PostgreSQL database using the schemas below. In offline scenarios, the system gracefully defaults to a local JSON structure.

1. Users table (gym_users)

Column	Type	Description
id	TEXT (PK)	Unique identifier for accounts
name	TEXT	Account holder name
email	TEXT (Unique)	Login credential email
password_hash	TEXT	Salted PBKDF2 hash of user password
tier	TEXT	Membership tier (Standard / Platinum / VIP)
joined	TEXT	Join date string (e.g. "Jun 2026")
streak	INTEGER	Active workout log streak counter
water_intake	INTEGER	Today's logged water intake (ml)
role	TEXT	System access role (member , trainer , admin)
avatar	TEXT	URL link to user profile photo

2. Classes table (gym_classes)

Column	Type	Description
id	TEXT (PK)	Unique class ID
class_name	TEXT	Name of group class session
trainer_name	TEXT	Staff member conducting the class
time	TEXT	Schedule time slot string
day	TEXT	Recurrence days
spots	INTEGER	Active slots remaining
max_spots	INTEGER	Total capacity of the class



Security & Payment Configuration

1. JSON Web Token (JWT) Security

Authentication is handled using native crypto tools without external packages:

- **Token Signature:** Utilizes HMAC-SHA256 with a `JWT_SECRET` key to construct secure browser cookies and authorization headers.
- **Header Authorization:** Requests to private endpoints must append `Authorization: Bearer <token>` to bypass server verification checks.

2. Stripe Checkout Integration

- **Checkout Generation:** Calling `POST /api/gym/checkout` returns a secure Stripe hosted url redirecting customers to complete payments in cards or UPI.
 - **Auto-Validation:** The webhook/success page retrieves the Stripe checkout session ID, asserts payment status is `paid`, decreases class spots, registers the class/trainer reservation, logs the transaction ledger, and dispatches a digital email receipt automatically.
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How to Export this Report to PDF

To share this specification document as a PDF:

1. Open the [gym_app_documentation.md](#) file inside VS Code.
2. Install the **Markdown PDF** extension.
3. Right-click anywhere in the file and select **Markdown PDF: Export (pdf)**.
4. Alternatively, open the markdown file preview, right-click, select **Print**, and set destination to **Save as PDF**.